Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-S-CLASS-P0171
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz S-Class (W223 / S 500 4MATIC)
Powertrain: 3.0L Turbocharged Inline-6 (M256) with 48V Integrated Starter-Generator (ISG) & Electric Auxiliary Compressor (eZV)
Engine Control System: Bosch MG1CP002 (Engine Control Module N3/10) - FlexRay / CAN-FD Gateway
Electrical Architecture: Dual 12V / 48V Mild-Hybrid Architecture
Fuel Injection Architecture: 200-350 bar Direct Injection (Piezo HP Fuel Injectors, dual HP fuel pumps)
Firing Order: 1-5-3-6-2-4

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0171`
- **Standard Fault Description:** System Too Lean (Bank 1)
- **System Definition:** Short-term and long-term fuel trim values on Bank 1 have exceeded the upper stoichiometric enrichment threshold (+25%).
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz S-Class (W223 / S 500 4MATIC)**:

1. **Crankcase Ventilation (PCV) Oil Separator Diaphragm Split:** Integrated inside M256 composite cam cover, drawing unmetered air directly into intake plenum.
2. **Charge Air Cooler (Water-to-Air) Coupler Seal Leak:** O-ring seal blow-out at cold-side charge pipe connection to throttle body (M16/6).
3. **48V Electric Booster (eZV) One-Way Check Flap Binding:** Air leak past recirculation gate causing intake manifold pressure transducer (B28/1) skew.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the S-Class control harness:

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[Sensor B28/1 Intake Manifold Pressure]     [ECM N3/10 Connector 'E']
Pin 1 (Sensor Ground) ----------------------> Pin 22 (Sensor Ground Ref)
Pin 2 (5.0V Reference Supply) <------------- Pin 23 (VCC 5.0V Regulated)
Pin 3 (Analog MAP Signal 0.5 - 4.5V) ------> Pin 24 (ADC Input Channel 4)

[EVAP Purge Valve Y58/1]
Circuit 87M (12V Battery Ign) ------------> Pin 1 (+12V)
Pin 58 (Low-Side PWM Ground Control) <----- Pin 2 (Switched PWM Ground)
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Electrical Reference Points

- **Primary Ground Reference:** `W10/4 (Cylinder Head Ground, Left) & W16/5 (Front Bulkhead Ground)`
- **Power Distribution Node:** `Front SAM (Signal Acquisition Module N10/6) Fuse Box F32/3`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Monitor Short Term Fuel Trim (STFT) and Long Term Fuel Trim (LTFT) in live data: observe if LTFT exceeds +22%.
2. Smoke test the intake tract via test port on clean-air intake duct at 0.5 bar pressure; inspect PCV breather hose and oil filler cap.
3. Clamp off the EVAP purge canister regeneration line to isolate fuel tank vapor influence.
4. Perform smoke test on intercooler bypass boots.

5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace integrated PCV / oil separator assembly (Part # A256010xxxx) on valve cover.
- Renew rubber charge air boot clamp and lubricate fluoroelastomer O-rings with Krytox.
- Reset mixture adaptation values via XENTRY under "Special Function -> Teach-in Processes".

6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.